

Supplemental Materials

An Empirical Study of Quality of Change-level Feature in JIT-SDP

Abstract—This document presents additional information of the submitted paper “Enhancing Feature Quality in JIT-SDP: An Empirical Study of Branch Dependency Mining”.

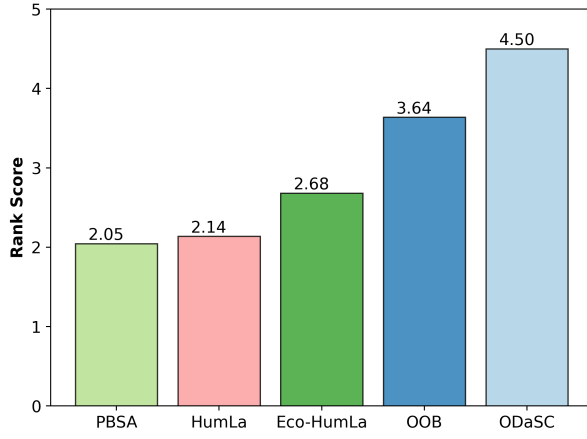
I. RQ3: IMPACT OF FEATURE NOISE ON MODEL RANKING

We have evaluated the rank score of the 5 models both for the two datasets (F^- and F^+) and used box plots to display and analyze the kendall tau [1] in our paper. Here we will display more detailed information about our results of the rank score. As shown in Fig 1, we compared the final Rank scores of the five models on F^- and F^+ both for G-Mean and MCC in our paper. The following context will display other metrics: Recall-0, Recall-1 and F1-measure.

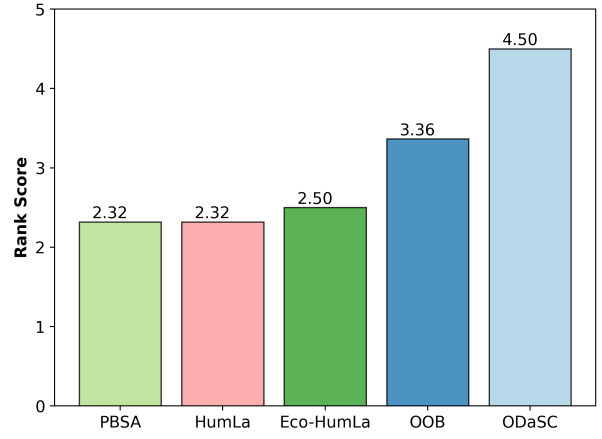
The results indicate that, in terms of specific numerical values, the rank score has indeed been affected, and we have observed some changes in the arrangement of certain models.

REFERENCES

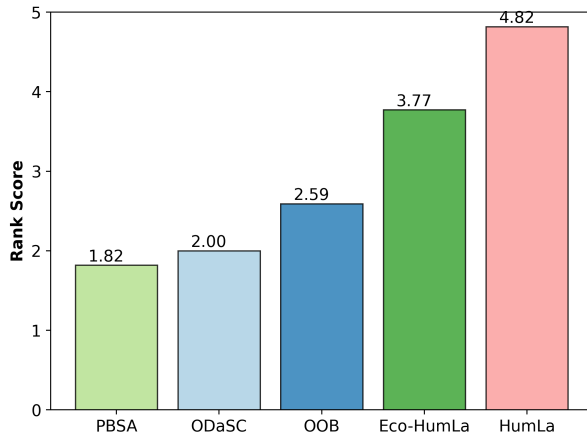
- [1] M. G. Kendall, “A new measure of rank correlation,” *Biometrika*, vol. 30, no. 1-2, pp. 81–93, 1938.



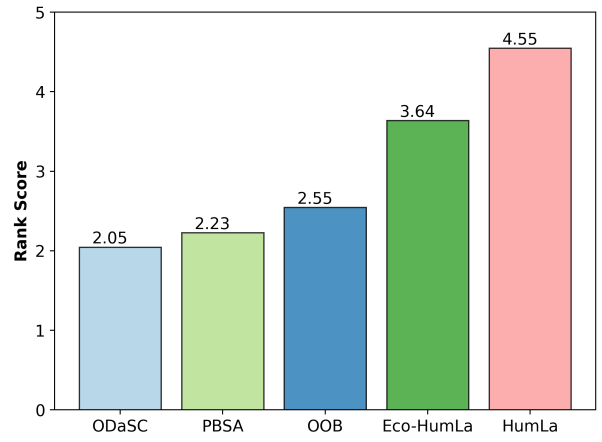
(a) Recall-0 Rank Score on noisy feature values F^-



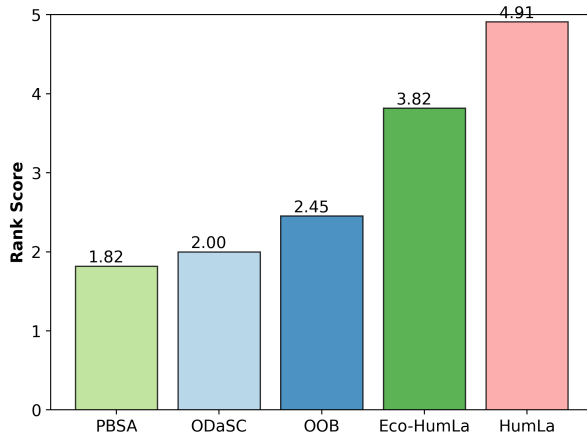
(b) Recall-0 Rank Score on new feature values F



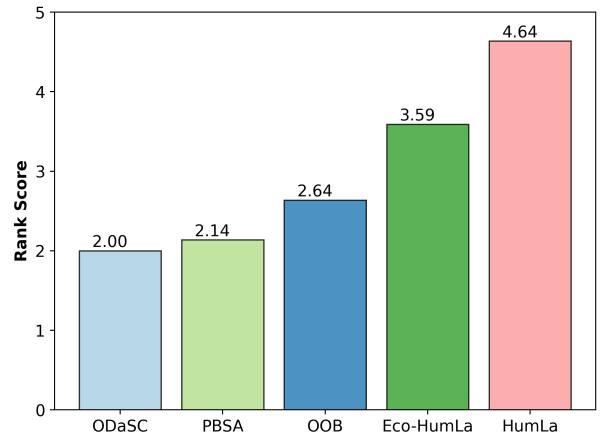
(c) Recall-1 Rank Score on noisy feature values F^-



(d) Recall-1 Rank Score on new feature values F



(e) F1-Score Rank Score on noisy feature values F^-



(f) F1-Score Rank Score on new feature values F

Fig. 1: Extra Rank Score for 5 Models